Game Documentation - SpaceWar

Student Information:

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1. Summary of the Game Concept

SpaceWar is a fast-paced 2D side-scrolling survival action game set in deep space. The player controls a high-tech starship navigating through a dense asteroid field while fending off enemy drone ships. The goal is to survive as long as possible, destroy enemies, and achieve a high score while managing health through power-ups and dodging obstacles.

2. Core Features Implementation

- **Movement System:** The player can move vertically and horizontally using conventional WASD or Arrow keys. The movement is bounded within the camera view to prevent the ship from leaving the play area.
- **Health System:** Implemented using a robust PlayerHealth script. The player has a maximum of 100 health points. Health decreases upon collision with enemies or asteroids and can be restored by collecting green Health Packs.
- **Scoring System:** Players earn points by destroying enemies and successfully passing obstacles. The score is displayed in real-time on the UI.
- **Enemy Spawning:** Enemies spawn at random intervals from the right side of the screen and move toward the player at varying speeds.
- **Projectile Combat:** The player can fire energy projectiles to destroy oncoming enemies.

3. Control Guide

- **Move Up:** W or Up Arrow
- **Move Down:** S or Down Arrow
- **Move Left:** A or Left Arrow
- **Move Right:** D or Right Arrow
- **Shoot:** Space Bar

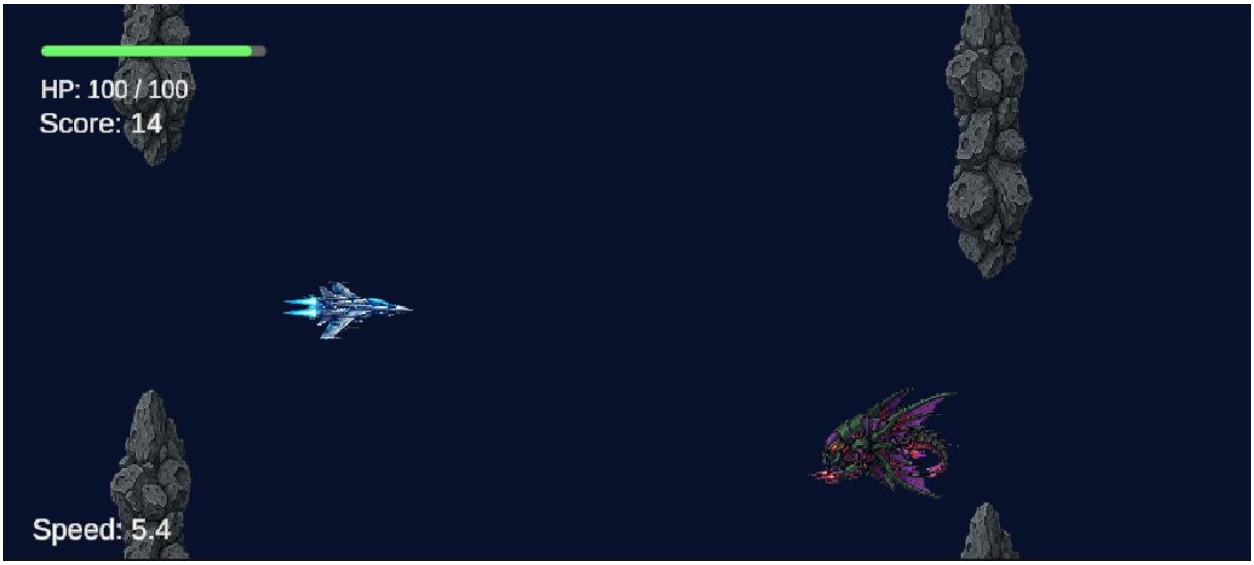
4. Creative Feature: Dynamic Difficulty & Visuals

The game features a **Dynamic Difficulty Scaling** system. As the game progresses, the scroll speed of the background and the spawn frequency of obstacles/enemies increase, making the game more challenging over time. Additionally, the **Asteroid Generation System** uses random scaling and rotation to provide a unique visual experience in every playthrough.

5. Gameplay Screenshots

- **Screenshot 1:**

- **Screenshot 2:**



- **Screenshot 3:**

6. Video Demo Link

- **URL:** https://drive.google.com/drive/folders/1mINo6bmujbQCtCKsj4uSz8Kq4IIbcxYl?usp=drive_link